

Rage Vs Reason: Analyzing Emotional Outbursts from Constructive Criticisms in Negative Steam Reviews

1st Diksha Anand

Dept. of ECE
NIAMT Ranchi
ce4112023@niamt.ac.in

2nd Chetan Mudlapur

Dept. of CSE
IIIT Bhagalpur
chetan.240201008@iiitbh.ac.in

3rd Om Prakash Singh

Dept. of CSE
IIIT Bhagalpur
opsingh.cse@iiitbh.ac.in

Abstract—Sentiment analysis generally classifies textual data into positive, negative, or neutral categories; however, the negative sentiment is mostly treated as a single, uniform category. In real-world context, negative reviews may contain either emotion-based reviews or constructive, reason-based responses. Our work addresses this limitation by introducing a fine-grained classification technique, which distinguishes rage-driven emotional responses from reasonable feedback within negative user reviews of games. A custom manually annotated dataset of 1001 game reviews collected from *Overwatch*® 2, *Kerbal Space Program 2*, and *Starfield* was manually annotated for this work. A traditional supervised machine learning model, support vector machine, was implemented and evaluated, along with a zero-shot and a few-shot large language model. The results showed that traditional supervised classifiers outperformed the other models with an overall accuracy of 82.09%. Additional statistical analysis of average word and character counts was performed to understand the structural differences between rage-driven and reason-driven texts. The proposed work provides a practical approach to assist developers in identifying constructive insights underlying in the negative reviews, which may help them improve the products and services.

Index Terms—Natural language processing, machine learning, large language model, SVM, text analysis.

I. INTRODUCTION

Sentiment analysis and emotion detection have become important research areas in Natural Language Processing (NLP), especially due to the rapid growth of user-generated textual content on various digital platforms. Online game reviews provide rich emotional expressions and opinionated data that reflect users' satisfaction, contentment, frustration, expectations, and reasoning about their experiences [1]. In this research, we worked on a dataset of game reviews collected from the games: *Overwatch*® 2 (Blizzard Entertainment, 2023), *Kerbal Space Program 2* (Intercept Games, 2023), and *Starfield* (Bethesda Game Studios, 2023). Based on the research of Dr. Paul Ekman, the seven standard, universal emotions which are expressed similarly regardless of culture are anger, contempt, disgust, fear, happiness (or enjoyment), sadness, and surprise [2]. Usually, when an emotion classification task is conducted,

the provided samples or reviews are classified into these emotions or into positive, negative and neutral [3]. But negative reviews can also contain some reasonable review or constructive thoughts which may eventually help in the betterment of such products [4]. These negative sentiments in the data are often treated as a single category. However, negative expressions may come from either emotionally driven reactions or constructive criticism. This paper addresses this gap by classifying negative text further into rage-driven emotional responses and reason-driven constructive feedback. This classification enables a fine understanding of users' intentions in feedback analysis. By separating emotional outbursts from constructive reasoning.

The main motive of this project is to highlight the underlying meaning in negative comments of users, which is usually neglected or overlooked. Negative Feedback does not always contain rage, anger or rudeness, but it may also include some useful, reasonable, constructive thoughts, some genuine feedback behind those negative reviews or ratings, which should reach to the developers or the managers so that they work on the product and service improvement [5]. We primarily wanted to address that "Is it possible to distinguish rage-driven responses from reason-driven constructive feedback within negative user reviews?"

The key contributions of this rage vs. reason research work are as follows: 1) Novel dataset creation: A custom, manually annotated dataset of 1001 negative reviews of the three selected games was created, labelled as rage or reason. 2) Comparative model evaluation: Multiple models were implemented and evaluated to compare their effectiveness in rage vs. reason classification, where the Support Vector Machine (SVM) model outperformed the other models. 3) Large Language Model (LLM) performance analysis: The study provides a comparison between traditional machine learning models and zero-shot, few-shot LLM techniques. 4) Additional Analysis: An additional statistical analysis was conducted between rage-driven and reason-driven texts by calculating their average word count and character count.

II. LITERATURE REVIEW

Nandwani and Verma [2] perform sentiment analysis and emotion detection from textual data to determine the human's attitude toward an item. The significance lies in implementing sentiment analysis along with an emotion detection model to determine an individual's mental state precisely. The paper consists of the implementation of lexicon-based, machine learning-based, deep learning-based and transfer learning approaches, along with data preprocessing and feature extraction methods. The deep learning approach and lexicon-based approach perform better, but there are limitations such as lack of resources, sarcasm & irony detection and lack of generalization.

Singh and Jaiswal [6] fully review the sentiment analysis concept and machine-learning algorithms across movie, product, and Twitter reviews. Their significance lies in comparing various ML and deep learning methods, datasets, tools, methods and performance metrics, and its application lies in business, politics and public actions. Methodologies implemented are supervised, unsupervised, and hybrid approaches used in prior studies. The result mentions that SVM gives the highest accuracy in comparison to other models. The limitations include domain-specific applications, multilingual analysis and limited real-time analysis.

Adam and Setiawan [7] perform sentiment analysis on the data which is taken from Twitter in Bahasa Indonesia, and to classify them into positive, negative, and neutral classes. The significance lies in building a system using Convolutional Neural Network (CNN) and Gated Recurrent Unit (GRU) methods. Methodologies implemented are preprocessing, feature extraction & expansion, Synthetic Minority Over-sampling Technique (SMOTE) and classification algorithms like CNN and GRU. Results show that CNN-GRU with fastText+SMOTE achieves the highest accuracy of 97.77%. Limitations include the dependency on labelled data, domain-specific tweets, and reduced performance due to unbalanced datasets.

Ramdan et al. [8] use sentiment analysis to classify film reviews into positive and negative sentiments and compare Logistic Regression (LR) and Support Vector Classification (SVC) models. Their significance lies in evaluating machine learning models for sentiment analysis so that it can be helpful in decision making process, whether to watch a film or not. Methodologies used are Term Frequency-Inverse Document Frequency (TF-IDF) for feature extraction, 70:30 train-test split on 2000 reviews, and evaluation of confusion metrics using LR and SVC algorithms. Results show that SVC performs better with 82.42% accuracy, whereas the LR model obtains an accuracy of 80.61%. Limitations include limited dataset, and lack of use of neural networks methods such as CNN, Recurrent Neural Networks (RNNs) & Long Short-Term Memory (LSTM).

Paspallis and Pannayiotou [9] aimed to develop an

intelligent web filtering mechanism in the form of a web browser extension. Their significance lies in developing a prototype extension to detect the sentiment of the articles and improve decision-making process. Methodologies implemented are preprocessing, metrics evaluation and machine learning classifiers such as SVM, LR and Bernoulli & multinomial Naive Bayes. Results show 79% highest accuracy using LR model to perform sentiment analysis. Limitations include difficulty in handling implicit aspects as there was no option to learn why an item was blurred.

Reddy et al. [10] compared the performance of BERT and RoBERTa models for sentiment analysis using the Steam Reviews dataset. The study highlights that the transformer models effectively handle large-scale, informal, and sarcastic gaming reviews. Used the Hugging Face Transformers library to fine-tune pre-trained BERT and RoBERTa models for the sentiment analysis task on a subset of 6.4 million reviews using an 80:10:10 train-validation-test split. RoBERTa model outperformed BERT model by achieving F1-score of 0.95. The difference in performance between BERT and RoBERTa models is not statistically significant.

Alfarisyi et al. [11] focused on sentiment analysis using Word2Vec and SVM to evaluates the impact of lemmatization and various SVM kernels for analyzing Steam RPG game review. The study aims to assist users by summarising reviews through sentiment results. The methodologies implemented are like preprocessing, tokenization, lemmatization, Word2Vec with SVM classifiers using a dataset of 10,000 user reviews. Achieved an 88.6% F1-score using 300-dimensional vectors and polynomial SVM kernels. Further research to be done on preprocessing aspects, such as stemming, word normalization, and data cleaning.

III. METHODOLOGY

Our work follows a systematic approach to analyse rage-oriented negative user reviews on three video game titles, namely *Overwatch*® 2, *Kerbal Space Program* 2, and *Starfield*, using both traditional machine learning model and LLM models. We selected titles with majorly negative player reception (40%+ negative reviews) on Steam and the exact quantification is shown in Table I. The methodology flow diagram is illustrated in Fig. 1. First of all, the dataset was collected for three games from official Steam API. The Python libraries such as *requests*, *pandas* and *tqdm* were applied to the dataset for efficient data retrieval, processing, and progress monitoring through extensible progress bars. After loading the complete user review dataset with 13,030 total reviews, only negative reviews were kept (7860 reviews) for further processing, as the objective of this study is to focus on rageful criticism and constructive, reasonable feedback. After the filtration, a total of 1001 data from negative reviews were manually annotated. Each review was classified into one of the two categories: rage, representing emotions, aggressiveness and reason, representing logical

reason or constructive criticism. Manual annotation helped to accurately interpret the review's intent, as sometimes using automated techniques fails to understand the nuance of human language and may falsely classify it as a rageful comment even when there is reasonable feedback. The criteria for "rage" annotation were those reviews that offered no feedback regarding game systems, game design, or any other tangible aspect of the game which clearly aid the developers by informing what areas to work on, rather expressing raw emotions via sarcasm, memes, or emotional negative outbursts. The criteria for "reason" annotation were those reviews that criticized aspects of a game in an informative manner, even if harsh tone was employed for expressing it, such that developers understand where the product falls behind, what features are not up to the community's expectations, and what functionalities to prioritize.

Once the dataset was refined with minimal pre-processing, three different models were applied to the annotated dataset. The labelled dataset was divided into training (X_{train}) and testing (X_{test}) sets, in 80:20 ratio via *train_test_split*, to train and evaluate model performance. The stratify parameter ensured equal proportion of classes in the training and testing sets as the original dataset. The TF-IDF vectorizer was used to convert textual reviews into numerical representations. First, a SVM classifier was implemented. Second, the Zero-Shot approach was utilized to evaluate the model's ability to classify reviews without any prior knowledge about the dataset. Finally, a Few-shot learning approach was used, where a limited number of labelled examples were provided to the model. The same X_{test} data was used across all three models to compare their performances across accuracy, precision, recall and F1 score. An additional word count analysis was performed for both classes.

TABLE I: Selected Games with High Proportion of Negative Reviews on Steam as of January 2026.

Game Titles	Reception on Steam	% Positive Reviews	Number of Reviews (English)
Starfield	Mixed	57%	83,062
Kerbal Space Program 2	Mostly Negative	26%	21,561
Overwatch® 2	Mostly Negative	34%	136,619

IV. IMPLEMENTATION

This section describes how all the models, namely SVM, Zero-shot and Few-shot, were implemented and executed, from dataset preparation to final evaluation. The custom dataset collected from official Steam API contains 13,030 user reviews for the three video games, *Overwatch® 2*, *Kerbal Space Program 2* and *Starfield* and the negative reviews were separated, over which manual

annotation is done over 1001 data, classifying the user reviews into one of the two categories: rage or reason, based on the emotion based and constructive, logical-based comments. The manual annotation ensures that the human emotions and nuance are correctly interpreted as automated techniques often fail to capture the underlying emotions of the reviews. After the balanced dataset (501 rage reviews, and 500 reason reviews) is constructed authentically using Python 3 libraries such as *pandas*, *requests*, and *tqdm* for data retrieval, cleaning, and progress bars, the dataset is applied to a traditional machine learning algorithm, SVM classifier and LLM techniques such as Zero-shot and Few-shot.

A. SVM Model

SVM model is implemented in the Central Processing Unit (CPU) runtime environment, and Python 3 libraries such as *numpy*, *pandas*, *scikit-learn* are used. Before applying the model text preprocessing is done in order to remove the noisy data so a custom text cleaning function was implemented for this purpose. At first, all characters in the text were converted into lowercase to prevent the model from treating the same word. (e.g. "Good" and "good") in different cases as separate features. Then, Uniform Resource Locator (URLs) present in the text were removed using regular expressions, as they don't contribute much meaning. After that, extra whitespaces such as multiple spaces or tabs were normalized into a single space, and other spaces were removed. Finally, the cleaned text was applied to the entire dataset using a column-wise transformation. This pre-processing step improves the quality of textual features and enhances the performance of machine learning models. The preprocessed dataset is then split into a train and test set in 80:20 ratio while maintaining a balance of rage v/s reason classification in both sets, using the stratified sampling technique. To convert the textual data into numerical features, the Term Frequency-Inverse Document Frequency (TF-IDF) vectorization technique was employed. The vectorizer was fitted on the training dataset and the same transformation was applied to the test dataset to ensure consistency and prevent data leakage. The test data was saved separately in a Comma-Separated Values (CSV) file so that it could be used later with a LLM. For classification, a linear SVM classifier was used, setting the parameters as $C=1.0$ and $random_state=42$ to maintain a balance between correct classification and model generalization. The trained SVM model learns to separate the rage and reason classes and is later used to make predictions on unseen test data. After that the accuracy of the model was calculated along with the classification report containing precision, recall, f1-score, and support. Then, with the help of matplotlib, confusion matrix for the SVM was plotted representing the true label and predicted label. After model evaluation, a basic text analysis was performed by calculating the character count. The average character and word counts were calculated separately for the



Fig. 1: Methodology Flow Diagram.

'Reason' and 'Rage' classes. These implementation helped in understanding how the two classes vary in terms of text length then, a histogram of character counts was plotted for both classes. The visualisation shows the character count distribution and spread of review lengths across both categories.

B. LLM Zero-Shot

At first we set up the runtime environment to T4 GPU to improve the speed and installed required libraries such as transformer to access pre-trained models, Pytorch for deep learning and -q flag is used to keep the installation output clean. A zero-shot classification pipeline was initialized using the pre-trained *facebook/bart-large-mnli* model from the Transformers library. This model enables text classification without pre-trained language understanding. This setup allows direct classification of unseen text into predefined labels such as Rage vs Reason. A custom function was defined to perform zero-shot classification using the LLM. For each input review, the model evaluates the text against predefined labels using a hypothesis template and selects the most suitable label. The predicted label is then mapped to binary values, where "Rage" encoded as 1 and "Reason" as 0. The progress of classification was displayed using a progress bar to handle the large amount of data efficiently. After that, the accuracy of the model was calculated along with the classification report containing precision, recall, f1-score, and support, and finally, the confusion matrix was plotted using the seaborn and matplotlib libraries.

C. LLM Few-Shot

After the Zero-shot LLM model, we implemented the few-shot model, and we provided limited labelled training data to the model, which helps it better understand the classification between rage and reason. While zero-shot classification relies only on the model's pre-trained knowledge, few-shot learning provides task-specific context, often leading to more accurate predictions. The pipeline was imported from transformers. The *google/flan-t5-large* model was used in a text-to-text generation setup, where classification is performed by prompting the model with example input reviews. A custom function was built to generate predictions by analyzing the model's output text and mapping it to binary labels, with "Rage" encoded as 1 and "Reason" as 0. The progress of classification was displayed using a progress bar using *tqdm* library. After that, the accuracy of the model was calculated along with the classification report and the confusion matrix was plotted using the seaborn and matplotlib libraries for the few shot model as well.

All three models were performed on a manually annotated custom dataset and were evaluated using the same input text data to ensure a fair comparison of performance.

V. RESULTS

The results obtained are: SVM model achieved an accuracy of 82.08%, Zero-shot model achieved 59%, and Few-shot model achieved 54%. Each model's performance was evaluated using a classification report that includes accuracy, precision, recall, and F1-score, as shown in Table II. A confusion matrix was plotted using the matplotlib and seaborn libraries, as shown in Fig. 2.

For the Reason class, the model achieved a precision of 0.81, a recall of 0.84, and an F1-score of 0.82. The recall value is high, which shows that the SVM model was effective in correctly identifying most reasonable reviews. For the Rage class, the model achieved a precision of 0.84, a recall of 0.80, and an F1-score of 0.82. The lower recall shows that some rageful reviews were misclassified as reasonable. The balanced macro and weighted F1-scores of 0.82 demonstrate that the model performs well for both classes without any bias. The confusion matrix (Fig. 2a) observed is that, out of 100 reasonable reviews, 84 were classified correctly, whereas 16 were misclassified as Rage. Similarly, 81 out of 101 Rageful reviews were correctly predicted, with 20 instances incorrectly labelled as Reason. This indicates moderate confusion in some cases where emotional tone and reasoning coexist.

In contrast, the LLM models could not perform that well because training data and fine-tuning resources for emotion-specific classification were limited. Since the SVM model was trained on the manually annotated custom dataset, it learned better and performed well. In contrast, the zero-shot model had no prior data, and the few-shot model used some data to train. The zero-shot approach performed better with 59% accuracy, in comparison to few-shot approach, with 54% accuracy, as *facebook/bart-large-mnli* is specifically pre-trained for Natural Language Inference (NLI). While the LLM techniques surpassed SVM in correctly predicting 'Rage' class, as is evident by Fig. 2b and Fig. 2c, the confusion matrix highlights the inability of LLM models to identify reviews with constructive criticism. These results suggest that, for limited resources, emotion classification tasks, traditional supervised learning models may perform better than the LLMs.

An additional analysis was conducted to examine the average character count and average word count for both rageful and reasonable reviews. The character count distribution was plotted, the x-axis contained characters, and the y-axis contained frequency. The total average

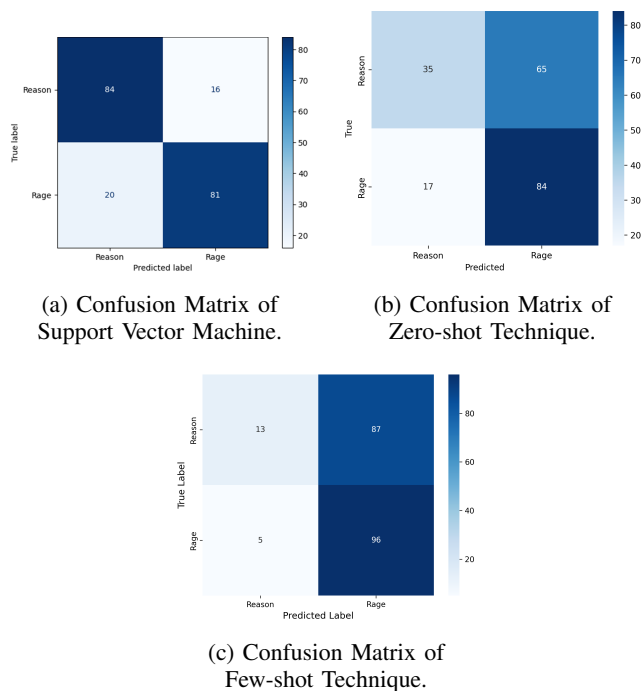


Fig. 2: Confusion Matrices.

TABLE II: Performance Comparison of Proposed Techniques.

Model	Accuracy	Macro Precision	Macro Recall	Macro F1 Score
SVM	82%	82%	82%	82%
Zero-shot	59%	62%	59%	57%
Few-shot	54%	62%	54%	45%

character count is as follows for reason(0): 1105.04 and for rage(1): 183.44. Similarly, the average word count was for reason (0): 197.75 and for rage (1): 33.81. From Fig. 3 it was concluded that reason-driven reviews have a higher average word count and average character count, with upto 8000 characters in some instances, indicating more detailed explanations and structured responses. In contrast, rage-driven texts were observed to be shorter and more aggressive in majority reviews, indicating impulsive emotional expression.

VI. CHALLENGES

The major challenges observed in the research are as follows: 1) The manual annotation of the rageful and reasonable texts. 2) Some of the reviews contain mixed feelings, such as sarcasm and implicit tone, which makes it difficult to label them. 3) Emotional expression varies significantly across different cultural styles and regional language influences. 4) The model may struggle to accurately interpret rage or reason in culturally nuanced or multilingual expressions. 5) Another factor that was a

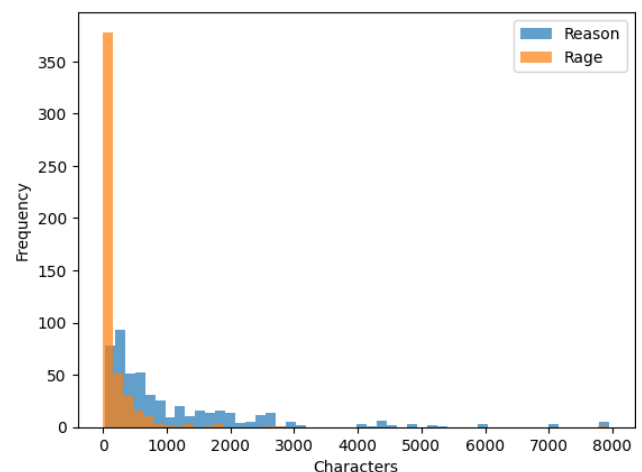


Fig. 3: Character Frequency in Rage and Reason Reviews.

major challenge was the bias and human dependency, as the manual annotation may be highly dependent on personal interpretation of the texts.

VII. CONCLUSION

This research work addressed an important gap in general sentiment analysis, that negative user feedback is not always a sudden outburst of emotion, but may also contain constructive and reasoned criticism. Traditional approaches usually treat negative sentiment as a single category, which limits their usefulness for developers. So to bridge this gap, we proposed the rage vs reason classification work to distinguish rage-driven emotional responses from reason-driven constructive feedback within the negative reviews. A custom dataset of game reviews was manually annotated for this classification. Three different techniques such as SVM, zero-shot and few-shot model were implemented and evaluated, where the SVM, i.e. supervised machine learning technique, outperformed the other two techniques.

The future work may incorporate the following: 1) Larger datasets: Increasing the size of the dataset may improve model generalisation and outcomes as there will be more data to train the model. 2) Multi-annotator labelling: Multiple human annotators must be present to perform the task to remove any kind of bias and personal opinions. 3) Machine models: Application of different techniques or models to improve the accuracy and fine-tune the classification. 4) Deep learning models: Exploring transformer-based models (e.g., BERT, RoBERTa), fine-tuned specifically for rage vs reason classification to capture deeper underlying meanings from the texts. 5) Real-time feedback analysis: Integrating the given framework into real-time review monitoring systems to help the developers in identifying meaningful feedback promptly.

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